

# Revised Load Interconnection System Impact Study Report

## For Queue Project Q12-0180

Indian Lake 138 kV  
August 30, 2014

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0180. The Interconnection Customer proposes a 115.9 MW large load (data center) facility interconnecting at the Indian Lake 138 kV substation in the Virginia area. The request was received on November 28, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 124 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,135,000.

### 3. Network Impacts

#### 3.1 Load Deliverability Analysis

| Monitored Facility                                                | Contingency                             | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------------------------|-----------------------------------------|-------------|--------------|----------|
| American – Aes Warrior Run Cogeneration Facility Substation 138kV | American – Ridgeley 345kV               | 180.1       | 430          | 5.9      |
| American 345/138kV TX                                             | Ridgeley – American 345kV               | 176.8       | 437          | 5.9      |
| Gravel Ridge – Shepherdstown 138kV                                | Urban Substation – American 345kV line  | 176.7       | 298          | 6.2      |
| Gravel Ridge – Dam No. 4 Hydro Station 138kV Line                 | American – Urban 345kV                  | 174.7       | 304          | 6.2      |
| Sleepy Hollow – Grimmway-Malaga 138kV                             | Kearneyville – Clark County 345kV       | 168.6       | 251          | 9.4      |
| Millville – Sleepy Hollow 138kV line                              | Clark County – Kearneyville 345kV       | 168.0       | 245          | 9.4      |
| Doglas City – Millville 138kV                                     | Keareyville – Clark County 345kV        | 166.1       | 242          | 9.4      |
| Aes Warrior Run Cogeneration Facility – Ridgeley 138kV            | American – Ridgeley 345kV               | 154.5       | 620          | 6.1      |
| Alligator Substation 500/345 kV Auto                              | Santee Cooper – Greenlandgap 345kV Line | 143.4       | 544          | 5.7      |
| Gene – Grimmway-Malaga Substation 138kV                           | Gallegos – Killingsworth 138kV Line     | 133.8       | 204          | 7.7      |
| Tanner – Lcec No. 5 138kV                                         | Gallegos – Killingsworth 138kV          | 132.2       | 221          | 7.7      |
| Ridgeley – American 345kV                                         | E Devon – Thiele Kaolin Reedy Crk 345kV | 131.9       | 1,263        | 11.2     |
| Tanner – Gene 138kV                                               | Gallegos – Killingsworth 138kV Line     | 131.2       | 217          | 7.7      |
| Lcec No. 5 – Shepherdstown 138kV                                  | Urban – American 345kV                  | 130.9       | 224          | 6.5      |

| Monitored Facility                                     | Contingency                                    | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------------|------------------------------------------------|-------------|--------------|----------|
| Huntley – Twin Lakes Alte 138kV                        | Twin Lakes Alte – Wpcp 138kV                   | 123.6       | 247          | 5.2      |
| International Paper Valliant Ok – Urban 345kV L        | Frontroyal – Urban 345kV                       | 122.0       | 1,276        | 11.2     |
| Gallegos – Killingsworth 138kV                         | Tanner – Lcec No. 5 138kV                      | 119.6       | 257          | 7.5      |
| Gallegos – Redman 138kV L                              | Tanner – Lcec No. 5 138kV                      | 118.3       | 253          | 7.5      |
| Redman – Glenbrook 138kV                               | Lcec No. 5 – Tanner 138kV Line                 | 117.5       | 249          | 7.5      |
| Ridgeley – Spindle Substation 345kV                    | Tiele Kaolin Reedy Crk – East Devon 345kV line | 116.7       | 1,495        | 11.0     |
| Mcconnico – Huntley 138kV                              | Twin Lakes Alte – Wpcp 138kV                   | 116.5       | 267          | 5.2      |
| Glenbrook – Palmyra 138kV                              | Tanner – Lcec No. 5 138kV L                    | 115.6       | 244          | 7.5      |
| Arvin – Southwest Research Institute 138kV             | Douglas City – Garden Corners 138kV            | 115.5       | 170          | 8.4      |
| Palmyra – Grimmway-Malaga 138kV                        | Tanner Substation – Lcec No. 5 138kV           | 113.1       | 241          | 7.5      |
| Southwest Research Institute – Douglas City 138kV Line | Garden Corners – Douglas City 138kV            | 112.9       | 201          | 8.4      |

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                 | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Aes Warrior Run Cogeneration Facility – American 138kV L | Rebuild line with higher-capacity conductor | \$2,750,000   | 8% (\$217,000)        |
| American 345/138kV TX                                    | Replace transformer with higher-rated unit  | \$11,069,000  | 8% (\$873,000)        |
| Gravel Ridge – Shepherdstown 138kV L                     | Rebuild line with higher-capacity conductor | \$1,562,000   | 6% (\$102,000)        |
| Gravel Ridge – Dam No. 4 Hydro Station 138kV Line        | Rebuild line with higher-capacity conductor | \$6,588,000   | 6% (\$430,000)        |
| Sleepy Hollow – Grimmway-Malaga 138kV                    | Rebuild line with higher-capacity conductor | \$1,952,000   | 21% (\$407,000)       |
| Millville – Sleepy Hollow 138kV line                     | Rebuild line with higher-capacity conductor | \$3,294,000   | 21% (\$686,000)       |
| Doglas City – Millville 138kV                            | Rebuild line with higher-capacity conductor | \$12,729,000  | 21% (\$2,651,000)     |
| Aes Warrior Run Cogeneration Facility – Ridgeley 138kV   | Rebuild line with higher-capacity conductor | \$7,443,000   | 4% (\$333,000)        |
| Alligator Substation 500/345 kV Auto                     | Replace transformer with higher-rated unit  | \$35,111,000  | 4% (\$1,312,000)      |
| Gene – Grimmway-Malaga Substation 138kV                  | Rebuild line with higher-capacity conductor | \$5,836,000   | 26% (\$1,501,000)     |

| Facility                                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Tanner – Lcec No. 5 138kV                                        | Rebuild line with higher-capacity conductor | \$9,589,000   | 26% (\$2,467,000)     |
| American Substation – Ridgeley 345kV L                           | Rebuild line with higher-capacity conductor | \$28,454,000  | 3% (\$947,000)        |
| Tanner – Gene 138kV L                                            | Rebuild line with higher-capacity conductor | \$733,000     | 26% (\$189,000)       |
| Shepherdstown – Lcec No. 5 138kV                                 | Rebuild line with higher-capacity conductor | \$3,756,000   | 11% (\$421,000)       |
| Huntley – Twin Lakes Alte 138kV                                  | Rebuild line with higher-capacity conductor | \$19,756,000  | 7% (\$1,385,000)      |
| International Paper Valliant Ok – Urban 345kV                    | Rebuild line with higher-capacity conductor | \$39,742,000  | 3% (\$1,254,000)      |
| Gallegos – Killingsworth 138kV line                              | Rebuild line with higher-capacity conductor | \$3,777,000   | 20% (\$764,000)       |
| Gallegos – Redman 138kV                                          | Rebuild line with higher-capacity conductor | \$2,349,000   | 20% (\$475,000)       |
| Redman – Glenrook 138kV line                                     | Rebuild line with higher-capacity conductor | \$2,142,000   | 20% (\$433,000)       |
| Ridgeley – Spindle Substation 345kV                              | Rebuild line with higher-capacity conductor | \$56,866,000  | 3% (\$1,831,000)      |
| Mconnico – Huntley 138kV                                         | Rebuild line with higher-capacity conductor | \$6,423,000   | 7% (\$450,000)        |
| Glenbrook – Palmyra 138kV                                        | Rebuild line with higher-capacity conductor | \$1,860,000   | 20% (\$376,000)       |
| Arvin Substation – Southwest Research Institute Substation 138kV | Rebuild line with higher-capacity conductor | \$11,068,000  | 23% (\$2,506,000)     |
| Palmyra – Grimmway-Malaga 138kV                                  | Rebuild line with higher-capacity conductor | \$2,265,000   | 20% (\$458,000)       |
| Southwest Research Institute – Douglas City 138kV                | Rebuild line with higher-capacity conductor | \$2,944,000   | 23% (\$667,000)       |
|                                                                  | Total allocated to project                  |               | \$23,135,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.